



The Spatial Economics Toolbox for Fisheries (FishSET) is an R package for modeling fisher behavior and simulating policy scenarios

Paul Carvalho^{1,2}, Lisa Pfeiffer^{2,*}, Anna Abelman^{1,3}, Min-Yang Lee⁴, Alan Haynie⁵

¹Pacific States Marine Fisheries Commission, Portland, OR 97219, United States

²Northwest Fisheries Science Center, NOAA Fisheries, Seattle, WA 98112, United States

³Alaska Fisheries Science Center, NOAA Fisheries, Seattle, WA 98115, United States

⁴Northeast Fisheries Science Center, NOAA Fisheries, Woods Hole, MA 02543, United States

⁵International Council for the Exploration of the Sea (ICES) Secretariat, Copenhagen 1553, Denmark

*Corresponding author. Northwest Fisheries Science Center, NOAA Fisheries, 2725 Montlake Blvd. East, Seattle, WA 98112 United States. E-mail: lisa.pfeiffer@noaa.gov

Abstract

The Spatial Economics Toolbox for Fisheries (FishSET) is an R package for end-to-end economic policy analysis of fishery management measures using discrete choice models. FishSET is a novel toolbox that allows users to organize and visualize fisheries-dependent data, apply cutting edge econometric models of location choice, and simulate the effects of alternative policy scenarios to understand the redistribution of fishing effort and welfare consequences of those policies. FishSET helps researchers overcome a variety of challenges inherent to this modeling framework, including standardizing data management, guiding model selection, and organizing statistical code to facilitate transparency and innovation. Fisheries management involves complex decision-making in a changing environment; policy objectives can be achieved more effectively with information about the potential impacts of proposed regulations. In fisheries, regulations often take the form of spatial or temporal management (i.e. restricting fishing within an area or a particular time period). FishSET is a decision-support tool designed to integrate fishery-dependent data, improving the analysis of fisheries management measures before implementation. FishSET offers an extensive set of models and model performance evaluations. FishSET facilitates the analysis of potential impacts on fisheries under various scenarios, including fishing ground closures resulting from regulations; new ocean uses, such as marine protected areas, offshore aquaculture, or wind energy projects; changes in fish stock distribution; and significant events, such as hurricanes or oil spills.

Introduction

Fisheries management employs scientific methods to limit the mortality of targeted and incidentally caught fish species; this supports the long-term sustainability of wild-capture fish stocks. At its core, fisheries management involves regulating the behavior of commercial and/or recreational fishers to meet these management objectives. Understanding fisher behavior is important for designing effective fishery management policy (Peterson 2000, Wilen et al. 2002, Hilborn 2007, Abbott and Haynie 2012, Fenichel et al. 2013, Abbott and Fenichel 2013, Vasquez Caballero et al. 2023). Since the 1980s, fisheries economists have used the Random Utility Model (RUM) framework, which is grounded in microeconomic theory, to model the choices of fishing location to understand spatial effort patterns and fishery participation from fishers' preferences revealed from fishery-dependent data (Bockstael and Opaluch 1983, Eales and Wilen 1986, Parsons and Kealy 1992, Dupont 1993, Ward and Sutinen 1994). These models are often empirically applied as discrete choice models, and are used

to understand the drivers of the decision-making process, estimate underlying behavioral parameters, and predict how fishers may respond to a wide variety of policies and other external drivers. Spatially explicit policies for fisheries management, such as area closures for bycatch reduction and marine planning policies, whereby policymakers wish to understand the trade-offs involved with allocating ocean space to distinct uses, are well-suited to analysis with discrete choice models because they can produce policy-relevant estimates of the redistribution of fishing effort and the welfare impacts of policy decisions (Hicks et al. 2004, Valcic 2009, Abbott and Haynie 2012, Girardin et al. 2017, Reimer et al. 2017, Dépalle et al. 2020). Here, "welfare" is a general term for a group's level of well-being, or utility. Utility is not directly measurable, so these models provide the means to represent changes in welfare in empirically measurable, monetary terms. Economic valuation estimates are necessary for cost-benefit analyses and useful for the incorporation of human communities into ecosystem-based management (Holland et al. 2010).

The Spatial Economics Toolbox for Fisheries (FishSET) is an R package (R Core Team 2024) for end-to-end economic policy analysis using discrete choice models applied to fisheries data. More general choice modeling software is available through commercial (e.g. Stata, SAS) and freeware packages (e.g. R, Hess and Palma 2019). FishSET allows users to organize and visualize fisheries-dependent data, apply econometric models of location choice, and simulate the effects of alternative policy scenarios to understand redistribution of fishing effort and welfare consequences of those policies. FishSET is designed to link elements of the marine environment with elements of human behavior through a transparent, replicable, and accessible framework for estimating a class of statistical models that have broad recognition in the field of fisheries economics and policy. FishSET is particularly suited to analyses that involve spatially explicit scenarios such as area closures: for example, from the establishment of marine protected areas (MPAs), energy extraction or transmission, shipping routes, or fisheries-management related area closures. FishSET can also be used to analyze the effects of species range shifts, changes in fishing costs or prices, or the implementation of other types of fishery management policies.

This paper describes the goals of FishSET, the framework used to organize the toolbox, and the details of the graphical user interface (GUI) developed for non-technical users. We also provide a replicable proof-of-concept for the toolbox that illustrates how FishSET can be used to analyze the effect of a spatial closure. The proof-of-concept uses the anonymized dataset included with FishSET to explore the potential effects of offshore energy development-driven spatial closures on the Atlantic Sea Scallop fishery. In the Northeast US, offshore energy projects overlap somewhat with the Atlantic Sea Scallop fishery and displacement has been an ongoing concern (Chaji et al. 2025). While the spatial closures are hypothetical (none of the cells are under consideration for closure), the workflow is available in a vignette and demonstrates the steps that a user may take with their own data and modeling question.

Goals of FishSET

FishSET was developed to achieve the following goals:

- Standardize data management and organization to facilitate the estimation of discrete location choice models with commonly available fishery data.
- Provide easily accessible tools to enable location choice models to better understand factors that drive fisher behavior and to provide input to the management of fisheries.
- Organize statistical code so that predictions of fisher behavior developed by the field's leading innovators can be incorporated and transparent to all users.

Fishery-dependent data [data produced from the activities of fishers, such as logbooks, delivery tickets, vessel monitoring system (VMS) data, or data from fisheries observers] can be challenging to access and manage. Such data are often confidential, as they contain personally identifiable information and/or privileged business information. Access to non-aggregated, confidential data is generally limited to staff at government agencies working directly on fisheries management. FishSET allows users to utilize state-

of-the-art data integration, visualization, and modeling tools locally, without any need to upload confidential or sensitive data. The data integration and visualizations tools are designed to guide users through the quality assurance/quality control (QA/QC) and data organization needed to use the models included in FishSET.

FishSET aims to make the analysis of a distinct set of policy questions using the RUM framework straightforward and achievable by a broad set of analysts, ranging from those with no programming experience using the FishSET GUI, to fisheries economists who can develop innovative likelihood functions that advance FishSET's capabilities. FishSET fills a need to better incorporate fisher behavior into marine ecosystem modeling, and advances the priorities of many fisheries and ocean management organizations around the world (ICES 2021, Goethel et al. 2023, FAO 2025).

Finally, FishSET provides an organized codebase to compare model outputs using different assumptions, model inputs, likelihood functions, and/or data processing steps. Academic publishing favors complex modeling advances, and the data processing and assumptions are rarely detailed in journal articles, making comparisons with the existing models difficult (Abbott and Wilen 2011, Dépalle et al. 2021). FishSET promotes extensive model comparison, reports goodness-of-fit, automates the analysis of predictive skill, and supports the inclusion of this material in appendices and/or on-line repositories.

FishSET framework

FishSET overview

The FishSET R package is a suite of tools and functions developed in the R statistical programming language (R Core Team 2024). FishSET is publicly accessible and can be downloaded from the GitHub repository: <https://github.com/noaa-nwfsc/FishSET>. This article refers to FishSET version 1.1.0, but the GitHub repository hosts the most recent version of the package. Functions included in the package focus on six key areas: data management, data analysis, data formatting, discrete choice modeling, policy simulation, and reproducibility. The following sections provide a brief description of the six key areas, and more detailed information on each section is available in the FishSET user manual (<https://noaa-nwfsc.github.io/FishSET/>).

Data management

Effective data management ensures consistency, improves efficiency, and supports reproducible results. To facilitate these goals, FishSET provides functions that help organize and manage data within structured project folders stored locally on the user's machine. When FishSET is installed, datasets are saved in a single location within an SQLite database. Files added to FishSET project remain local and private. FishSET preserves an unaltered copy of the "original" data as a table in the database. A separate "working" table is created automatically, used for analysis, and can be modified by other FishSET functions. For a more detailed description of the folder structure within a FishSET project (e.g. SQLite database, outputs, etc.), see the user manual available on the FishSET website.

Data analysis

Exploratory data analysis is a critical preliminary step that enables an analyst to gain insight about the quality of a dataset and formulate hypotheses. FishSET contains functionality to support exploratory data analysis. This includes the ability to assess the distribution, data sparsity, and variance of individual variables through intuitive tables and visualizations (including mapping functions). Functionality to examine correlations and check for linear relationships between variables is also provided. Fleet summary functions help the user identify meaningful groupings and assign distinct fleets (or *métiers*). These data analysis tools provide users with a solid foundation for developing robust discrete choice models.

Data formatting and processing

Formatting data for discrete choice models can be complex and time consuming because data must be structured in a format that reflects the fisher's actual choice *and* the available alternatives. FishSET streamlines this process by providing a range of data preparation functions. Simple tools include data modification functions such as assigning unique identifiers to trips or hauls, scaling variables, creating dummy variables to represent categorical variables (i.e. one-hot encoding), and calculating catch-per-unit-effort. In addition, FishSET offers more advanced data processing functions, including the ability to calculate expected catch or revenue using user defined moving averages and generate distance or travel cost matrices between fishing locations. These data formatting and processing functions allow users to efficiently implement a project with raw data and minimal preprocessing. Data can be formatted as trips (such as a single day trip) or hauls (trips with multiple hauls).

Discrete choice modeling

FishSET is designed to estimate parameters of discrete location choice models to better understand factors that drive fisher behavior and to predict how policy measures will change choices and welfare. FishSET features five common likelihood functions (described in more detail below) and flexibility in the model design. FishSET can be used with trip-level data (a single location choice) or haul-level data (multiple location choices within a trip, with each choice dependent upon the previous). Users can incorporate explanatory variables that either interact with distance from port (e.g. vessel size) or relate to fishing locations (e.g. sea surface temperature). FishSET facilitates rigorous model selection by enabling users to easily estimate multiple models, compare estimated coefficients and t-values across models, and compare model fit. In addition, the package includes k-fold cross-validation functionality to evaluate model robustness. Further implementation details of discrete choice modeling in FishSET are provided below.

Policy simulation

Once a preferred discrete choice model is formulated and fit, users can simulate the effects of alternative policies, such as the implementation of marine protected areas, to predict changes in the spatial distribution of fishing effort and resulting welfare gains or losses. In the future, we plan to make more policy options available for simulation (e.g. seasonal closures). FishSET provides users with a GUI that streamlines the design of these scenarios and

generates intuitive tables and figures that summarize simulation results.

Reproducibility

Function calls in FishSET are logged and saved in a file within the project directory. The report log of function calls saves the function name, input parameter values, and outputs. Function messages that provide simple and non-critical information, such as the presence of missing values, are also stored. Storing messages provides a record of the data processing steps, such as removing rows from the data that contain missing values. All steps throughout the course of a project are automatically saved and the analysis can be easily reproduced at any time and/or rerun with updated data.

Data requirements

All FishSET projects require two datasets: a “primary” dataset and a “spatial grid” file. The primary dataset serves as the core input for discrete choice modeling (see FishSET user manual for file format compatibility). This dataset must contain a unique identifier for each fishing trip or haul, the date which the fishing event occurred (time of day is optional), the catch and/or revenue, trip start and/or end location, and the fishing location. Each row of the primary dataset should represent a unique observation, or choice occasion, with each column representing a unique variable. Location data (trip start, fishing location) can be stored in coordinate form, or using an identification variable linked to a table of grid cells or ports loaded in the FishSET project. Additional information such as prices, species caught, and vessel characteristics may be included in the primary dataset.

The spatial grid file divides the marine area of interest into discrete sub-areas, referred to as “alternatives” in FishSET. The spatial grid file must contain the latitude and longitude coordinates that define the boundaries of location polygons, and each location must have a unique identifier. These polygons can represent a variety of spatial units or configurations, such as regular grids, hexagonal grids, or boundaries based on fishery management or regulatory areas. All rows from the primary dataset should be associated with exactly one grid cell. The spatial grid is also used to calculate area centroids, and generate distance matrices between the starting location and all alternatives, and to compute cell-specific metrics. Acceptable spatial file formats include shapefiles, JSON, GeoJSON, or R data files (e.g. *.rds*). In addition, the FishSET website includes a “Making a spatial grid file” vignette that guides users through the process of creating a custom grid to use with their data.

Optional datasets include port data, auxiliary data, and gridded data. If supplied, the port dataset should contain the coordinates (latitude and longitude) of ports used by vessels in the primary dataset and a port name or identifier that is the link to the primary dataset. Auxiliary data can contain any additional variables that can be linked to the primary dataset by a common variable. For example, if the primary dataset contains a vessel identifier, the auxiliary data could provide variables related to vessel characteristics. Gridded data contains one or more variables associated with locations in the spatial grid. It can also vary by a second dimension (e.g. gridded data may be daily or hourly observations of a variable associated with each spatial grid location). Gridded data can

be used as explanatory variables in discrete choice models, such as sea temperature, wave height, ice cover, wind speed, or the distribution of a fish species. Refer to the user manual for more examples and appropriate formatting for port, auxiliary, and gridded datasets.

Implementation of discrete choice models

Discrete choice random utility models have become widely used in fisheries (Smith 2005, Girardin et al. 2017, Dépalle et al. 2020). Commonly used econometric models include conditional multinomial logit models, nested logit models, and mixed logit models (Train 2002). Many model variants and formulations have been developed by economists in fisheries and other fields, with innovations motivated by the policy question being analyzed, the assumptions of each model, the nature and extent of available data, computing power increases, and the statistical background of the analysts (Ellison and Glaeser 1997, Revelt and Train 1998, Klier and McMillen 2008, Tra 2010, Krisztin et al. 2022). A larger number of model variants are available in the choice modeling software package Apollo (Hess and Palma 2019). Here we provide a brief overview of the model formulations in FishSET.

These models assume that n fishers choose, from a discrete set of j fishing locations, a fishing location where the utility derived from each alternative is U_{nj} , $j = 1, \dots, J$. The fisher chooses the alternative that results in the greatest utility, thus choosing alternative i if and only if $U_{ni} > U_{nj}$, $\forall j \neq i$. Utility is decomposed into aspects that the researcher can and cannot observe, $U_{nj} = V_{nj} + \varepsilon_{nj}$, where ε_{nj} is the unobservable random error and assumed to follow a Gumbel (Type I Generalized Extreme Value) distribution. For results to be interpreted in practical terms of willingness to pay, utility must include explanatory variables in monetary units such as profit, cost, or income. The observable component of utility V_{nj} is often modeled as a linear combination of expected catch or expected revenue, distance from the fisher's current location, information about the vessel or fisher, costs, and/or additional information about the potential fishing locations (e.g. depth, temperature, spatial quota limits) that may affect monetary-based measures of utility.

To estimate parameters of discrete choice models in FishSET, users must define the set of alternative fishing locations to be included in the model—see function documentation for `create_alternative_choice()`. This involves specifying a starting location. For trip-level data (where each row represents a unique trip), the starting location is typically the port of departure. For haul-level data (where each row represents a unique fishing event within a fishing trip), the starting location for the first haul would be the port of departure, and starting location for subsequent hauls would be the location of the previous haul. Once the appropriate inputs are defined and `create_alternative_choice()` is executed, FishSET will generate a distance matrix between the observed starting locations and all alternative fishing locations. In the FishSET GUI, users are provided with a histogram showing the number of observations per location to check for sparsity.

Conditional logit models also require an expected catch or revenue matrix, which represents the catch or revenue that a fisher would expect to find in each alternative fishing location. This is often calculated as a moving average of observed catch or revenue. The `create_expectations()` function in FishSET sim-

plifies this step and allows users to customize both the averaging window (in days) and the lag period (in days) before the window begins. Multiple catch or revenue matrices (e.g. with different averaging windows and/or lag periods) can be included in a single model, but users should assess potential collinearity when using this model structure. Model results can be quite sensitive to these assumptions, so it is recommended that this step be carefully considered and sensitivity analysis performed with respect to these assumptions. There is a large set of potential functional forms, types of data to include, and ways to weight data that can be used to form expectations (Abbott and Wilen 2011). For detailed guidance on using this function, see the documentation for `create_expectations()` on the FishSET website.

FishSET currently provides five statistical functions for maximum likelihood estimation of discrete choice models: conditional logit (McFadden 1974); zonal (or alternative specific constant) logit (Abbott and Wilen 2011); and expected profit models (EPMs) with normal, lognormal, and Weibull catch functions (Haynie and Layton 2010). The conditional logit and zonal logit differ by the specification of the distribution of unobserved factors. The expected profit model offers endogenous estimation of the expected catches.

For the conditional logit model, the probability of fishing a location for a given observations is:

$$p_{ij} = \frac{\exp\left(\sum_{m=1}^M \beta_m G_{ijm} + \sum_{n=1}^N \gamma_n T_{in} D_{ij}\right)}{\sum_{k=1}^K \exp\left(\sum_{m=1}^M \beta_m G_{ikm} + \sum_{n=1}^N \gamma_n T_{in} D_{ik}\right)}, \quad (1)$$

where p_{ij} is the probability of fishing alternative j for observation (choice occasion) i , K is the total number of alternatives in the choice set, and M is the total number of location-specific variables. Location-specific variables (G_{ikm}) are specific to each alternative, such as catch rates or sea surface temperature. Location-specific variables can be interacted with seasonal dummy variables to generate seasonal covariates for the model. Variables that interact with distance between the starting location and each alternative location (D_{ik}) are represented by T_{in} , and N is the total number of distance-interacting variables ($n = 1, 2, \dots, N$). For example, distance-interacting variables could be vessel characteristics that affect the level of disutility experienced when traveling greater distances to fishing locations. For the conditional logit model, a β_m parameter value for each location-specific variable (G_{ikm}) and a γ_n parameter value for each distance-interacting variable (T_{in}) are estimated. The conditional logit model relies on the assumption the unobserved factors are uncorrelated for all alternatives, and have the same variance for all alternatives (Train 2002). This assumption is called Independence from Irrelevant Alternatives (IIA).

The conditional logit and zonal logit have similar equations for the probability of fishing in a location. The zonal logit model replaces $\beta_m G_{ijm}$ from (1) with the term $\beta_{jm} G_{im}$. In the zonal logit, the G_{im} variables do not vary across alternative locations. For example, G_{im} might represent vessel gross tonnage (GT), thus β_j parameters capture the effect of changes in vessel GT on the utility of location j . The alternative-specific constants capture the average effect of factors that are not included in the model. The model estimates $K-1$ β_j parameters for each variable G because only the differences in these values matter, and the value of $\beta_{j=1}$ is normalized to zero.

Table 1 Models based in the RUM framework.

Model	Use cases	Citations*
<i>Currently included in FishSET</i>		
Conditional logit/multinomial logit	Independence of irrelevant alternatives (IIA)	Abbott and Wilen (2011), Depalle et al. (2020), Birkenbach et al. (2022)
Alternative specific constant logit	Captures the average impact on utility of all factors that are not included in a model in a constant for each alternative	Birkenbach et al. (2024) Backstrom and Woodward (2023)
Expected Profit Model	Allows endogenous estimation of expected catches, can handle discrete/continuous choices	Haynie and Layton (2010)
<i>Other RUM models not currently in FishSET</i>		
Mixed logit/random parameters logit/latent class/error components	Relax assumption of IIA by allowing for random taste variation, unrestricted substitution patterns, and correlation in unobserved factors over time or space	Revelt and Train (1998), Smith (2005), Haynie et al. (2009), Wu et al. (2025)
Poisson	Relax assumption of IIA; number of agents is not fixed	Guimarães et al. (2004)
Generalized extreme value/Nested logit	Relax assumption of IIA but can assume IIA holds within each nest; Allows estimation of an outside option	Vasquez Caballero et al. (2023), Hicks et al. (2020), Stafford (2018)
Discrete choice dynamic programming	Model optimal dynamic exploitation paths, path dependency, game theory	Hicks and Schnier (2006), Hicks and Schnier (2008), Huang and Smith (2014)

*Girardin et al. (2016) includes a table with this information for many publications prior to 2015.

For the EPM with a log-normal catch function, the probability of fishing location i for observation j is:

$$p_{ij} = \underbrace{\left(2\pi\sigma_j^2 Y_{ij}^2\right)^{-\frac{1}{2}} \exp\left(\frac{-[\ln(Y_{ij}) - \sum_m^M \beta_{jm} G_{im}]^2}{2\sigma_j^2}\right)}_i \times \underbrace{\left(\frac{\exp\left[\frac{P_i \exp\left(\sum_m^M \beta_{jm} G_{im} + \frac{\sigma_j^2}{2}\right) + \sum_n^N \gamma_n T_{in} D_{ij}}{\sigma_\varepsilon}\right]}{\sum_a^A \exp\left[\frac{P_a \exp\left(\sum_m^M \beta_{am} G_{im} + \frac{\sigma_a^2}{2}\right) + \sum_n^N \gamma_n T_{in} D_{ia}}{\sigma_\varepsilon}\right]}\right)}_{ii} \quad (2)$$

where part i of the equation represents the log-normal catch function (i.e. log-normal probability density function), and part ii represents the logit choice component. The expected, or mean, value in the log-normal probability density function is parameterized as the linear combination of β_{jm} and G_{im} , which represent alternative-specific parameters and choice occasion variables, respectively. The normal and Weibull versions of the EPM replace part i with corresponding density functions. Similar to the zonal logit model, G_{im} variables do not vary across alternative locations (e.g. vessel size), but these variables can affect catch differently across alternatives through the β_{jm} parameters. The standard deviation (σ_j) is constrained to positive values via an exponential function, and Y_{ij} is the actual catch. The expected mean catch func-

tion in part ii of the equation is:

$$\exp\left(\sum_m^M \beta_{jm} G_{im} + \frac{\sigma_j^2}{2}\right), \quad (3)$$

and multiplied by price P_i to calculate expected revenue. The cost portion of part ii is the same as the conditional and zonal logit models—the summed products of γ_n , T_{in} , and D_{ij} . The common scale parameter σ_ε is distributed as Type I Extreme Value and represents unobserved expected profit heterogeneity (Haynie and Layton 2010).

The FishSET model design function (`make_model_design`) streamlines the process of selecting a statistical model, adding explanatory variables, and selecting expected catch matrices (for conditional logit models). The `discretefish_subroutine` function runs the maximum likelihood estimation routine and allows users to estimate multiple models simultaneously. Upon successful optimization, FishSET saves and outputs estimated parameter values, standard errors, t-statistics, the inverse Hessian matrix (approximates the covariance matrix of the parameter estimates), and metrics of model fit (AIC, AICc, BIC, and Pseudo R^2).

More complex models that relax IIA or other assumptions are abundant in the academic literature. Table 1 provides a comparison of use cases and examples to help users determine which type of model might be most appropriate in their own context. Most of these models build upon each other, and FishSET provides the framework to build model complexity while comparing model performance with simpler models. The “Future Development” section describes models that will be added and the

flexible model design functionality that will make it straightforward for users to specify additional models. Other modeling methods outside of the RUM framework can also be applied to the problem of predicting the location choice of fishers, such as agent-based or machine learning methods (e.g. support vector machines, decision trees). However, FishSET is limited to the RUM framework because of the formal link between the microeconomic theoretical objective function and the likelihood function of the empirical model, which allows the interpretation of estimated parameters and the use of monetary covariates as the marginal utility of income for welfare analyses (Schmidheiny and Brühlhart 2011).

Model diagnostics

FishSET provides convergence indicators and diagnostics in the case of non-convergence. The model fitting function provides users with convergence codes, errors, and warning messages from the optimization function, the standard errors for coefficients, and the Hessian matrix. Convergence codes (code = “0” indicates successful convergence) and messages returned from the optimization function will alert users if the model fails to converge. If the standard errors are NaN (not a number), this indicates that the Hessian is singular because the Hessian cannot be inverted to calculate standard errors. Another sign that models are failing to converge is uncharacteristically large coefficient and/or standard error values. If the MLE fails to converge, users should examine data for variables with zero observations, variables that perfectly predict the chosen zone (“complete separation”), or collinearity. However, if the MLE seems to converge but the coefficients and standard error are large, users should examine variables that are always zero for the chosen zone (“quasi-complete separation”).

Model robustness

FishSET has a k-fold cross-validation function (`cross_validation`) which makes it easy to evaluate model robustness and predictive skill. All k-fold outputs (parameter estimates, standard errors, and metrics of model fit) are saved locally to the user's machine as dated files in the “output” folder for each FishSET project. Also, the % absolute prediction error (Klaiber and von Haefen 2019) is calculated for each test group, which quantifies the deviation of the model's predictions from the observed data for each test group (see section 8.9 in the FishSET User Manual for more details).

We encourage users to also explore model robustness to spatial resolution of grid size. Abbott and Wilen (2011) and Dépalle et al. (2021) demonstrate that coarse spatial resolution can introduce structural bias in discrete choice models, especially for spatially heterogeneous resources. Structural bias resulting from spatial resolution can be explored in FishSET by creating separate project folders with different spatial grid files and comparing the outcomes (McFadden pseudo- R^2 and model residuals). In the future, we plan to add features that would make it possible to conduct this analysis within a single FishSET project.

Users should test multiple logit models if possible to compare coefficient stability. Estimating a zonal logit model (with alternative-specific constants) can reveal systematic unobserved utility in the system (mean preference for specific locations). For example, if the model coefficients change drastically between the conditional logit and zonal logit, covariates in the conditional logit

could be serving as proxies for location affinity rather than “true” preference for those covariates—a case of omitted variable bias (Tardiff 1978, Train 2002). Even when the IIA assumption is violated for the conditional logit model, this robustness check provides a baseline and information that can be used to build more complex models (e.g. mixed logit).

Policy simulation

After models are developed and the parameters are estimated, FishSET can be used to analyze “policies,” which are broadly defined as any hypothetical or actual exogenous changes that may affect utility or reduce the set of alternatives. Policies can be simulated through changes in the model inputs. Examples of policy evaluations include area closures, the opening of new areas, changes in prices, changes in absolute or relative catch rates, or changes in total allowable catches (TAC). While FishSET is designed to simplify the evaluation of several common types of policies, any explanatory variable that is included in the original model can be changed and evaluated in a “policy evaluation” framework.

The two types of policies that are currently integrated into FishSET are 1) area closures, and 2) changes in location-specific TAC. Area closures can result from policies such as bycatch or protected species area management closures, closures to protect bottom habitat that limit any fishing or fishing by certain gear types, marine protected areas, or other forms of biodiversity preservation or enhancement. Area closures can also result from the use of ocean space by industries other than fishing, such as designated shipping lanes, aquaculture areas, offshore wind energy development areas, or oil and gas drilling. Fisheries may be excluded from these areas for safety reasons or because the applicable legal framework permits exclusive use. In addition, locations may no longer be attractive for fishing because of shifting water temperature envelopes or extirpation of target species. The second type of policy, changes in location-specific TAC can be used to simulate effort controls where fishing is allowed up to a certain amount in particular locations.

FishSET calculates two primary outputs which estimate the effects from a change in policy: 1) the redistribution of aggregate fishing effort, and 2) the welfare loss or gain resulting from changes in policy or changes in other factors that influence fisher location choice. Outputs from running the policy function are automatically saved in the project “output” folder. Tables, bar graphs, and maps can be generated to display the redistribution of fishing effort across locations (see proof-of-concept section below for examples).

Changes in welfare for logit models (e.g. conditional logit, zonal logit) are simulated as the difference in the expected surplus, which is the expected utility in dollars or any monetary units (Train 2002, Hicks et al. 2004):

$$W = \frac{1}{\theta} \left[\ln \left(\sum_a^A \exp[V_a] \right) - \ln \left(\sum_b^B \exp[V_b] \right) \right], \quad (4)$$

where θ represents the marginal utility of income, B and A refer to “before” and “after” the policy change, and V is the representative utility (denominator of the conditional and zonal logit likelihoods). Welfare changes should be interpreted as the net change in utility from the policy in question, considering the behavioral response (i.e. the redistribution of fishing effort) due to the policy.

If the utility is specified in terms of profit, the change in welfare can be interpreted as the net change in profit. Welfare gains can also be interpreted as the willingness to pay for the policy change, and welfare losses can be interpreted as willingness to accept a payment for the policy change.

Estimating the marginal utility of income θ requires a variable in monetary units (such as profit, cost or revenue) in the function of representative utility. The choice of variable depends on the observable data available and the specification of the utility function: if profit maximization is assumed then the best measure of profit should be used to proxy utility. In practice, the negative of the cost variable's coefficient is θ (or simply the value of the coefficient of the profit or revenue variable). For example, if β_1 represents cost in the utility model, then the marginal utility of income $\theta = -\beta_1$ because the negative of the cost coefficient represents the increase in utility for a monetary decrease in cost (Train 2002). Dividing the difference in utility between before and after policy change by θ (utility divided by monetary units) converts the output into monetary units (Train 2002), which makes it possible to estimate the welfare loss/gain from policy changes. See the FishSET user manual or Haynie and Layton (2010) for more details on calculating welfare loss/gain for EPMs

GUI description

The FishSET GUI is an R Shiny application that guides users through the process of estimating the parameter coefficients of discrete choice models and simulating policy scenarios (Fig. 1). Included with the installation of the FishSET package, the GUI can be launched with the `run_fishset_gui()` function. The GUI was designed to empower users, regardless of their programming background, to import data, construct models, and explore the behavioral responses of fishers under different policy conditions. By lowering the technical barrier, the FishSET GUI supports a broad range of users in leveraging advanced modeling tools to inform fisheries management and decision making. It also allows groups of analysts to share responsibilities of model development and policy evaluation.

The FishSET GUI is organized into five tabs: upload data, QA/QC, format data, models, and policy. The GUI is structured such that users are guided through the tabs in a linear fashion. Each session of the GUI starts with uploading data (new data or load data from an existing FishSET project) and selecting variables to use from the datasets. Users will then move to QA/QC (Fig. 1) and check for data quality issues, such as missing data or observations on land. The format data tab includes functions for defining the set of alternative fishing locations and generating expected catch or revenue matrices. The model and policy tabs allow users to design and run discrete choice models and policy scenarios. As described above, the FishSET GUI will save a copy of the original data in a local SQLite database, while a working table will be updated as users work through each tab. Additionally, any data visualizations generated in the GUI can be saved to the FishSET project folder. To ensure reproducibility, all function calls, including input arguments, are also saved to a file in the project folder.

The GUI and codebase functions in FishSET are fully integrated. Thus, users can seamlessly switch between interacting with FishSET through the GUI and running code within the same project. While the GUI eliminates the need to write code, it offers less flexibility. This tradeoff was necessary to enhance usability and im-

prove the user experience of the GUI. The capabilities of the GUI have been carefully selected to provide users with an understandable, linear workflow that achieves the main objectives of formatting a dataset, estimating model parameters, and conducting policy simulations. However, there are more advanced capabilities and customization options available through the codebase of FishSET. For many users, the GUI is an initial step for data management and preliminary model development before accessing the more flexible capabilities of the codebase.

Future development of FishSET

The current version of FishSET provides users with a framework to develop relatively simple discrete choice models. While these models help build a solid foundation, the IIA assumption should be explored in each case. We are currently developing a diagnostic function to run the Hausman–McFadden test to determine whether the IIA assumption is appropriate (Hausman and McFadden 1984). To complement the Hausman–McFadden test, we are also developing a function to calculate global Moran's I on the residuals of the logit model, which detects spatial correlation in the model's errors. If spatial correlation of the residuals exists, then the IIA assumption is violated.

As users develop intuition and an understanding of the assumptions of each model, users will likely want to implement more complex models that are better suited for real-world data (Table 1). To address this need, we are working on integrating more modeling options in FishSET such as nested logit, mixed logit (random parameters logit), and spatial logit (accounting for spatial autocorrelation) models (see Table 1). In addition, the code base of FishSET has a modular design such that users can develop their own models and request to have them integrated into the FishSET package (<https://noaa-nwfsc.github.io/FishSET/articles/likelihood-template.html>).

We are also currently redesigning the model fitting function in FishSET, which will estimate model parameters using the RTMB (R bindings for Template Model Builder) package (Kristensen 2025). RTMB uses automatic differentiation to calculate the gradient of the likelihood function (or the gradient of the Laplace Approximation for models with random effects) and compiles the code into C++, which significantly improves computational efficiency over standard logit model fitting functions in R. This update will extend the ability of FishSET to handle large datasets with thousands of alternative fishing locations.

Proof-of-concept for the U.S. Atlantic Sea scallop fishery

An example application of FishSET to the U.S. Atlantic Sea scallop fishery is included as a set of two vignettes that are available on the FishSET website. The “Scallop Example” vignette is an exploratory data analysis using FishSET. It demonstrates a range of FishSET functions for uploading data, performing data quality assessment/quality control, and summarizing and visualizing data. The “Scallop Conditional Logit Model Example” vignette explains how to design and estimate a discrete choice model in FishSET using the scallop dataset. In addition, we illustrate how FishSET can be used to assess the impacts of policy or ocean use changes by

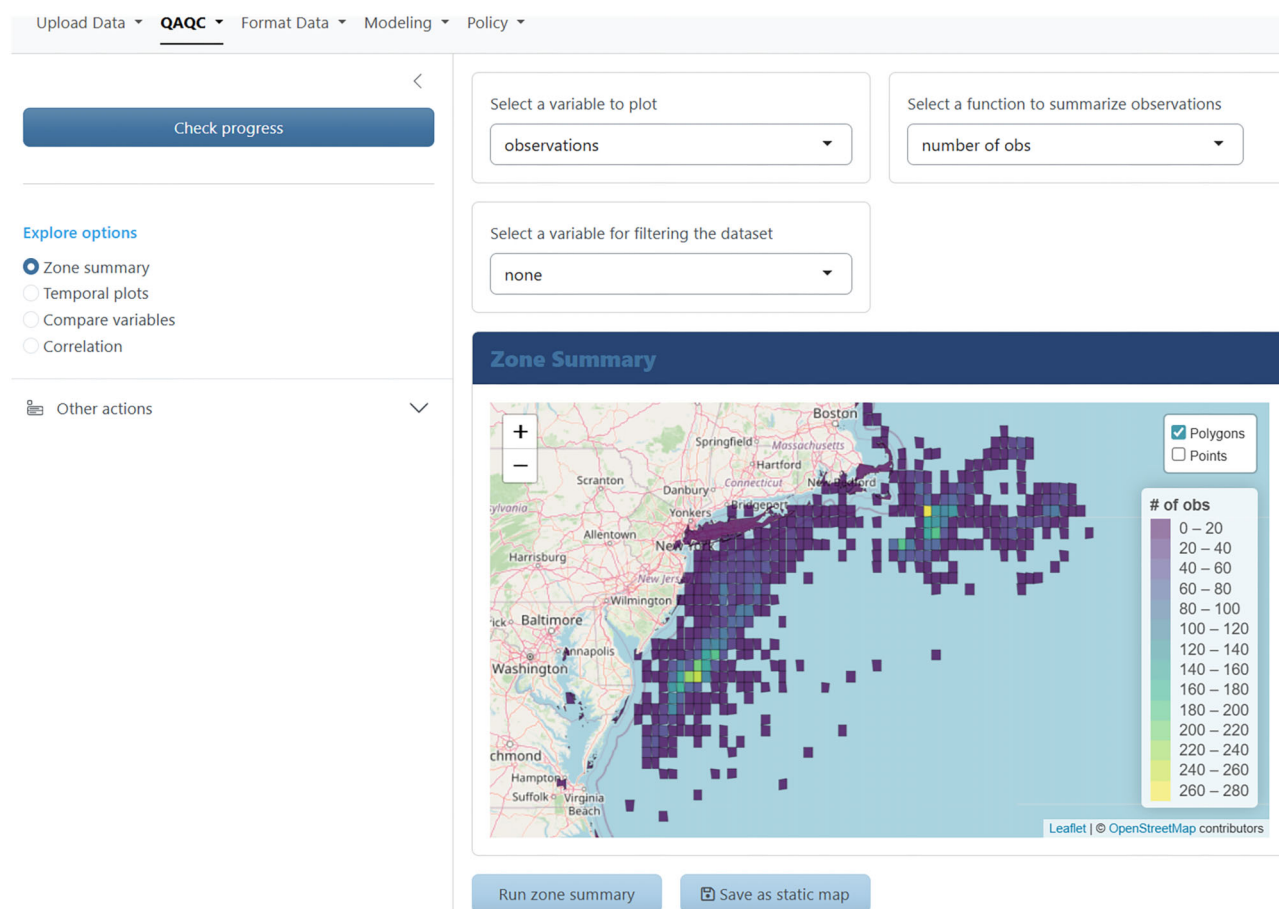


Figure 1 Screenshot of the FishSET GUI showing the zone summary function in the QAQC tab. Users are able to select variables to plot by fishing location (zone), plot different summary values (mean, median, sum, etc.), and filter the data by a categorical variable.

exploring the welfare impacts of the development of offshore wind energy on the sea scallop fishery.

The development of offshore wind energy off the U.S. East Coast has been a concern to regional fishing industries, including the scallop fleet. The construction of wind turbines involves local closures of ocean space to accommodate the construction. Operation of the wind farms does not necessarily mandate closures of the leased wind energy areas to fishing; however mobile bottom-tending gear, like scallop dredges and trawls, are likely to have difficulty operating near wind turbines and buried cables. Wind energy areas are likely to act as *de facto* closure areas for the scallop fleet and FishSET can be used to evaluate the welfare implications of spatial closures.

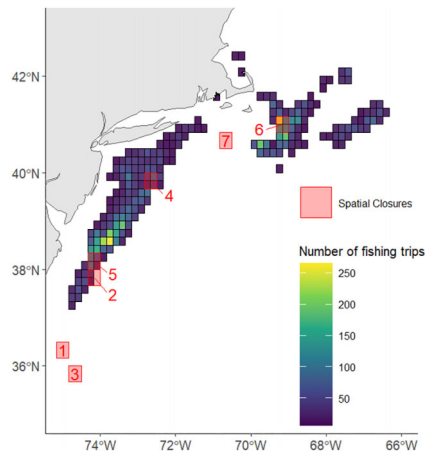
A sample dataset, containing anonymized data from the Limited Access Northeast United States Atlantic Sea Scallop (*Placopecten magellanicus*) fishery, is included with the FishSET package (FishSET::scallop) to help FishSET users gain intuition into required data structures, data elements, and processing capabilities of FishSET. The trip-level information in this sample dataset includes a unique identifier for each trip, the date and time of the beginning of the trip, catch, revenue, fishing location, start location, and end location. It also includes information about the nature of the fishing trip, including gear used, trip length, scallop landings, scallop revenue, revenue from all species, estimated costs, and a vessel identifier. The vignettes use a dataframe which contains a random sample of 10 000 trips from vessels

in the Limited Access Days-at-Sea fleet when they are declared into either an Access Area or Open area Days-at-Sea fishing trip (For more information on the commercial Atlantic Scallop fishery, see <https://www.fisheries.noaa.gov/species/atlantic-sea-scallop/commercial>). A small amount of random error (i.e. noise) was added to fishing locations, landing quantities, and the value of catch to obscure the original dataset and maintain confidentiality. Permit, operator, and trip identifiers were anonymized. Rows in this dataset contain trip-level information to estimate a basic location choice model with a few explanatory variables. The spatial grid (FishSET::tenMNSQR) is included with the FishSET package. The spatial grid defines a 10×10 min grid from 33° to 46° latitude and -77° to -64° longitude off the coast of the northeastern United States where the scallop fleets operate. An optional port dataset (FishSET::scallop_ports) is also included.

One of the fundamental steps in planning for a new ocean use such as offshore wind is to evaluate the potential exposure of existing users to the new ocean use. For fisheries, this involves first evaluating the degree of overlap of historic fishing activity with the areas identified for potential new uses. FishSET makes this task simple, as demonstrated by the sections in the “Scallop Example” vignette titled “Zone Summary” and “Closure Summary,” which generate figures and tables summarizing data for each fishing location and hypothetical wind lease areas that were created for this proof-of-concept (Fig. 2). These FishSET tools were used for the analysis in Chaji et al. (2025), which uses actual data from the At-

Table 2 Estimated parameters from “Scallop Conditional Logit Model Example” vignette.

	Estimated coefficient	Standard error	T-value
$\beta_{\text{expected catch}}$ (thousands of pounds)	0.093	0.001	97.793
$\beta_{\text{trip cost}}$ (\$thousands)	−0.056	0.001	−43.431

**Figure 2** Summary of hypothetical scallop fishing effort (number of trips to each location) and spatial closures (i.e. wind lease areas that serve as *de facto* spatial closures).

Atlantic Sea Scallop fishery and data from the actual wind energy lease area identification process to assess the potential economic exposure of the fishery to wind leases.

Simply calculating the overlap of the footprint of historical activities with that of possible new activities is insufficient for understanding the potential impact of the new activities (Dépalle et al. 2020). Fishers will change their fishing activity in response to a new set of conditions and constraints. In this example, we assume fishers choose the fishing location that maximizes their utility and that we are able to estimate model parameters that recover fishers' underlying behavior using past fishing data. Using the example scallop dataset and assuming a trip cost of \$100/mile traveled [See Werner et al. (2020) for empirical estimates of trip costs in this fishery], we estimate a simple conditional logit model with two explanatory variables: expected catch (a seven-day average of observed catch, lagged by one day) and travel cost. For real-world data, users should explore multiple lag settings for expected catch or revenue, explore including other observable variables, and compare model fits (Abbott and Wilen 2011, Girardin et al. 2017). The estimated parameters can be interpreted as the marginal change in the logged odds of taking a trip to any zone given a one-unit change in the independent variable (Train 2002). In this example, a 1000 lb increase in expected catch increases the log odds of choosing a fishing location by 0.093, or the odds of choosing a particular area are multiplied by $\exp(0.093) = 1.097$ for an additional 1000 lbs of expected catch, holding trip cost constant. Holding catch constant on the other hand, a \$1000 increase in the cost of a trip decreases the log odds of choosing a fishing location by 0.056 [or the odds multiplied by $\exp(-0.056) = 0.946$]. The willingness to pay in this example is \$1.66 ($-\beta_{\text{expected catch}}/\beta_{\text{trip cost}}$) for an additional pound of expected catch. Note that these results do

not represent actual estimates for the Atlantic Sea Scallop fishery given the stylized nature of the included data.

After fitting the model parameters, our function output indicates that the model converged successfully (convergence code = “0”), with a maximum gradient component $7e-5$ and a positive definite Hessian matrix (“pdHess” = TRUE, also confirmed by non-NaN standard error values). Next we can examine the model's theoretical validity (IIA assumption) by running a Hausman-McFadden test. Although this feature is currently under development for FishSET, users can run this as a chi-square test on the “full” model (with all alternative locations) and a “restricted” model (omitting one or more locations and refitting the model). For this example, we found that the IIA assumption is likely violated because there is a significant difference (P -value < 0.05) between the estimated parameter values for the “full” and “restricted” models. While this is to be expected for a conditional logit model on spatial data, our “restricted” model parameter estimates were $\beta_{\text{expected catch}} = 0.094$ and $\beta_{\text{trip cost}} = -0.056$. When the IIA assumption is violated, we still recommend running model robustness checks (e.g. running a zonal logit with ASCs) and estimating predictive power (e.g. k-fold cross validation) as a baseline for more complex models. For brevity, we will skip these steps for the proof-of-concept example.

Area closures are expected to have costs if there is overlap with the fisheries footprint (since fishers were maximizing utility prior to the closure). The results of the conditional logit can be used to compute the change in welfare (in monetary units) to estimate and compare the costs of potential closures. We simulated two area closures (areas 4 and 6) using the hypothetical wind areas depicted in Fig. 2. Using the estimated parameters in Table 2, we estimated the redistribution of fishing effort in response to closures (Fig. 3). We used the coefficient for trip cost (−0.056) to represent the marginal utility of income (θ in equation 4) to calculate the welfare impact. In FishSET, the redistributed fishing effort and welfare estimates are calculated using the `run_policy()` function.

As depicted in Fig. 3, effort was constrained to zero in the closed locations, and fishing vessels redistributed their effort to other locations. When area 4 was closed, the “hotspot” of fishing in the northeast area remained. However, when area 6 was closed the fishing effort was more evenly distributed in the large southwest area. The mean welfare loss per trip is estimated to be \$459/trip from a closure of area 4, and \$656/trip from a closure of area 6. The total welfare impact (across all trips by the vessels in the dataset) is estimated to be \$4 363 539 for the closure of area 4, and \$6 234 788 for the closure of area 6 (Table 3). Note again, however, these results are for demonstration only and estimated from a small anonymized subset of all scallop trips.

Conclusion

Policy makers are interested in the potential impacts of proposed regulations. In fisheries, these regulations often take the form of

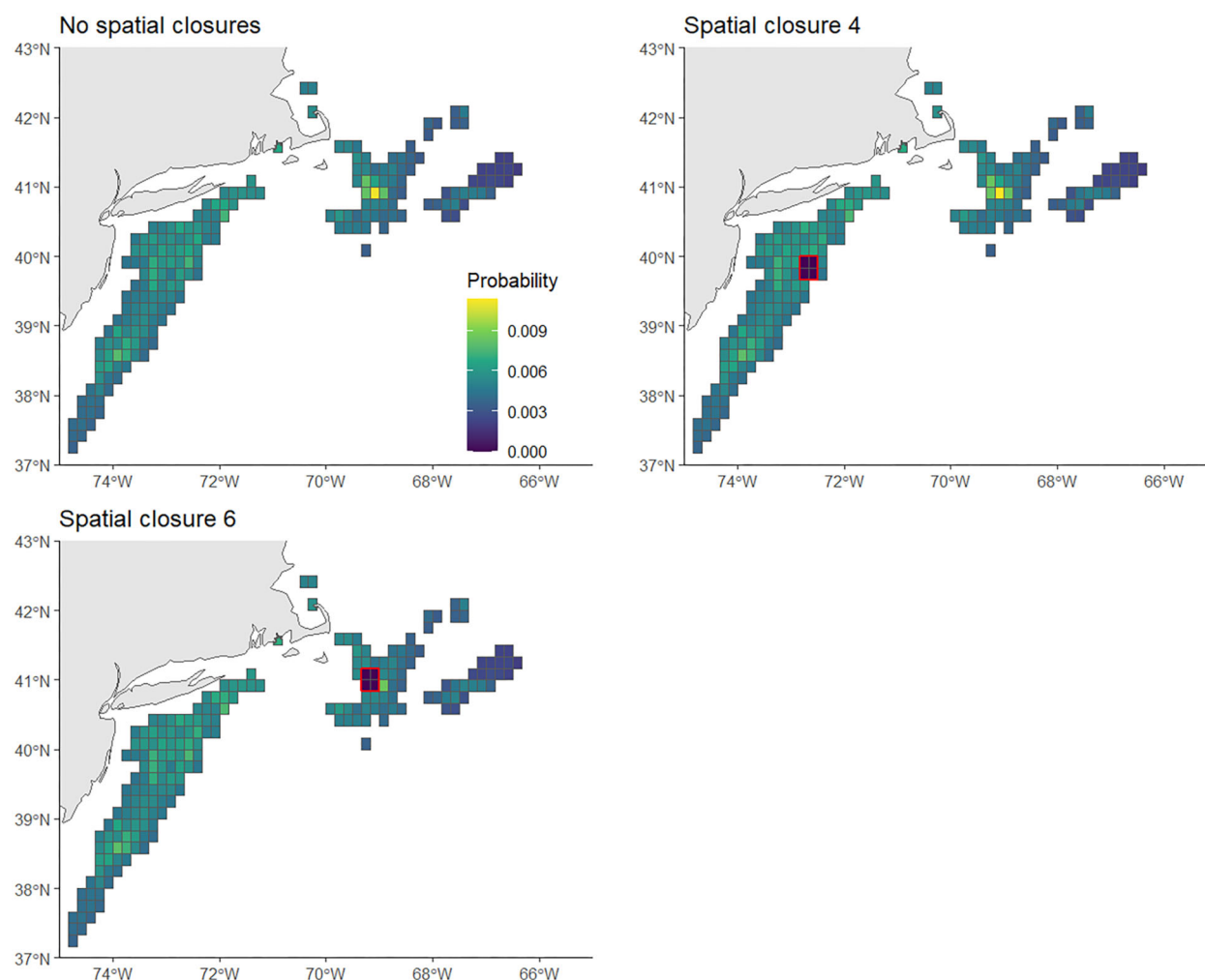


Figure 3 Predicted redistribution of fishing effort. Locations in the red squares indicate area closures.

Table 3 Estimated welfare impacts of two hypothetical spatial closure scenarios.

Scenario	Mean loss per trip (USD)	Total welfare loss (USD)
Spatial closure 4	\$459	\$4 363 539
Spatial closure 6	\$656	\$6 234 788

spatial or temporal management (i.e. restricting fishing within an area or during a particular period of time). Empirical work in fisheries has demonstrated that fishers respond to profit differentials over space, time, and across target fisheries by altering their choices about where, when, and what to fish. Fishers also respond to constraints by considering expected profit differentials in their alternative choices. Structural models that explicitly consider profit- or utility-maximizing behavior under relevant economic and institutional constraints are needed to provide reliable predictions of the impacts of proposed regulations (Reimer et al. 2022). These models capture the individual decision-making process within the salient institutional context, which may include economic factors such as prices and costs, environmen-

tal factors such as the spatial distribution of fish stocks, and management factors such as catch limits, area closures, or temporal regulations.

FishSET provides a user-friendly, approachable toolbox for developing such models and empirically applying them. FishSET includes a set of commonly-used models that can be used to estimate underlying behavioral parameters, such as the degree of by-catch avoidance (Curtis and Hicks 2000, Hicks and Schnier 2008, Abbott and Wilen 2011), the disutility of travel distance (Eales and Wilen 1986), risk avoidance (Pfeiffer and Gratz 2016), or habit formation (Holland and Sutinen 2000). They can also be used to predict the *ex ante* effects of a policy, or understand the drivers of observed changes after a policy is implemented (Smith and Wilen 2003, Abbott and Haynie 2012, Vasquez Caballero et al. 2023, Birkenbach et al. 2024). They can be used as a starting point to build additional model complexity. Finally, they can be used to examine the welfare impacts of policies or exogenous change (Baerenklau and Provencher 2005, Hunt et al. 2007, Haynie and Layton 2010, Dépalle et al. 2021). The ability to estimate welfare is an important aspect of this class of models, as it allows the direct, proper measure of economic benefits to society. By looking at before/after or predictive outcomes, one can measure the *change* in welfare due to the policy or exogenous change, thereby providing an important management indicator. FishSET is widely

relevant for analyzing changes in policy or other exogenous impacts on fisheries. Examples include the establishment of marine protected areas (MPAs), bycatch area closures, industrial exclusion areas, species range shifts, changes in fishing costs or fish prices that affect the differences in profitability and risk over time or space, or the implementation of various fisheries management actions such as catch share policies.

FishSET creates additional value by providing the functionality to examine a large number of data and modeling assumptions. Empirical work has demonstrated that complex relationships exist between spatial decisions and the specification of the spatial unit of analysis (Dépalle et al. 2021), the specification of expectations over space (Abbott and Wilen 2011), and the inclusion of participation nests (including non-fishing and/or fishery choice decisions) (Vasquez Caballero et al. 2023). It is challenging to determine the right set of assumptions for any given dataset and model. FishSET provides the codebase to iteratively build complexity, compare model outputs using different assumptions, models, and/or data processing steps, reports goodness-of-fit, automates the analysis of predictive skill, and supports the inclusion of this material in appendices and/or on-line repositories.

FishSET is an important tool for understanding and predicting the human response to a changing environment and the management measures implemented to protect it. It will allow fisheries and ocean management organizations and their stakeholders to better evaluate how management, economic, and environmental change impact fisheries and fishery-dependent communities. FishSET also facilitates improved data integration of economic data across regions that allows better coupling of economic, ecosystem, and stock assessment models, improving the sustainability and resilience of fisheries and fishery-dependent communities worldwide.

Acknowledgements

Thanks to Bryce McManus, Melanie Harsch, Corinne Bassin, Adam Hayes, Allen Chen, Rita Curtis, Doug Lipton, Marina Chaji, Jordan Watson, and Abigail Golden for their contributions to this project over the years. Thanks to three anonymous reviewers who provided valuable comments on the manuscript.

Author contribution

All authors contributed to the conceptualization, design and methodology, analysis, drafting, and revising the manuscript.

Conflicts of interest

The authors have no conflicts of interest to declare.

Funding

Funding was contributed by National Oceanic and Atmospheric Administration award number NA24NMF472C0018.

Data availability statement

The data underlying this article are available at <https://noaa-nwf-sc.github.io/FishSET/>.

References

- Abbott JK, Fenichel EP. Anticipating adaptation: a mechanistic approach for linking policy and stock status to recreational angler behavior. *Can J Fish Aquat Sci* 2013;**70**:1190–208. <https://doi.org/10.1139/cjfas-2012-0517>
- Abbott JK, Haynie AC. What are we protecting? Fisher behavior and the unintended consequences of spatial closures as a fishery management tool. *Ecol Appl* 2012;**22**:16. <https://doi.org/10.1890/11-1319.1>
- Abbott JK, Wilen JE. Dissecting the tragedy: a spatial model of behavior in the commons. *J Environ Econ Manag* 2011;**62**:386–401. <https://doi.org/10.1016/j.jeem.2011.07.001>
- Backstrom JD, Woodward RT. Onsite characteristics and diversity avoidance in marine recreational fishing demand. *Mar Resour Econ* 2023;**38**:435–63. <https://doi.org/10.1086/726025>
- Baerenklau KA, Provencher B. Static modeling of dynamic recreation behavior: implications for prediction and welfare estimation. *J Environ Econ Manag* 2005;**50**:617–36. <https://doi.org/10.1016/j.jeem.2005.03.001>
- Birkenbach AM, Cojocar AL, Smith MD et al. Discrete choice modeling of Fishers' Landing locations. *Mar Resour Econ* 2022;**37**:235–62. <https://doi.org/10.1086/719929>
- Birkenbach AM, Lee M-Y, Smith MD. Counterfactual modeling of multispecies fisheries outcomes under market-based regulation. *J Assoc Environ Resource Econ* 2024;**11**:755–96. <https://doi.org/10.1086/727356>
- Bockstael NE, Opaluch JJ. Discrete modelling of supply response under uncertainty: the case of the fishery. *J Environ Econ Manag* 1983;**10**:125–37. [https://doi.org/10.1016/0095-0696\(83\)90021-9](https://doi.org/10.1016/0095-0696(83)90021-9)
- Chaji M, Ardini G, Harsch M et al. Assessing impacts of offshore wind development: an analysis of the minimization of economic exposure of the scallop fishery through the regulatory process. *Fish Oceanogr* 2025 e12717. January 14, <https://doi.org/10.1111/fog.12717>
- Curtis R, Hicks RL. The cost of sea turtle preservation: the case of Hawaii's pelagic longliners. *Am J Agric Econ* 2000;**82**:1191–7. <https://doi.org/10.1111/0002-9092.00119>
- Dépalle M, Sanchirico JN, Thébaud O et al. Scale-dependency in discrete choice models: a fishery application. *J Environ Econ Manag* 2021;**105**:102388. <https://doi.org/10.1016/j.jeem.2020.102388>
- Dépalle M, Thébaud O, Sanchirico JN. Accounting for fleet heterogeneity in estimating the impacts of large-scale fishery closures. *Mar Resour Econ* 2020;**35**:361–78. <https://doi.org/10.1086/710514>
- Dupont DP. Price uncertainty, expectations formation and fishers' Location choices. *Mar Resour Econ* 1993;**8**:219–47. <https://doi.org/10.1086/mre.8.3.42629067>
- Eales J, Wilen JE. An examination of fishing location choice in the pink shrimp fishery. *Mar Resour Econ* 1986;**2**:331–51. <https://doi.org/10.1086/mre.2.4.42628909>

- Ellison G, Glaeser EL. Geographic concentration in US manufacturing industries: a dartboard approach. *J Polit Econ* 1997;**105**:889–927. <https://doi.org/10.1086/262098>
- FAO. Fisheries and the BBNJ Agreement. Food and Agriculture Organization. 1 December 2025, date last accessed. <https://openknowledge.fao.org/handle/20.500.14283/cd6494en>
- Fenichel EP, Abbott JK, Huang B. Modelling angler behaviour as a part of the management system: synthesizing a multi-disciplinary literature. *Fish Fish* 2013;**14**:137–57. <https://doi.org/10.1111/j.1467-2979.2012.00456.x>
- Girardin R, Hamon KG, Pinnegar J *et al*. Thirty years of fleet dynamics modelling using discrete-choice models: what have we learned? *Fish Fish* 2017;**18**:638–55. <https://doi.org/10.1111/faf.12194>
- Goethel DR, Omori KL, Punt AE *et al*. Oceans of plenty? Challenges, advancements, and future directions for the provision of evidence-based fisheries management advice. *Rev Fish Biol Fish* 2023;**33**:375–410. <https://doi.org/10.1007/s11160-022-09726-7>
- Guimarães P, Figueiredo O, Woodward D. Industrial location modeling: extending the random utility framework. *J Reg Sci* 2004;**44**:1–20. <https://doi.org/10.1111/j.1085-9489.2004.00325.x>
- Hausman J, McFadden D. Specification tests for the multinomial logit model. *Econometrica*, 1984 1219–40. <https://doi.org/10.2307/1910997>
- Haynie AC, Layton DF. An expected profit model for monetizing fishing location choices. *J Environ Econ Manag* 2010;**59**:165–76. <https://doi.org/10.1016/j.jeem.2009.11.001>
- Hess S, Palma D. Apollo: a flexible, powerful and customisable freeware package for choice model estimation and application. *J Choice Model* 2019;**32**:100170. <https://doi.org/10.1016/j.jocm.2019.100170>
- Hicks RL, Kirkley J, Strand IE. Short-run welfare losses from essential fish habitat designations for the Surfclam and Ocean Quahog fisheries. *Mar Resour Econ* 2004;**19**:113–29. <https://doi.org/10.1086/mre.19.1.42629421>
- Hicks RL, Schnier KE. Dynamic random utility modeling: a Monte Carlo analysis. *Am J Agric Econ* 2006;**88**:816–35. <https://doi.org/10.1111/j.1467-8276.2006.00900.x>
- Hicks RL, Schnier KE. Eco-labeling and dolphin avoidance: a dynamic model of tuna fishing in the Eastern Tropical Pacific. *J Environ Econ Manag* 2008;**56**:103–16. <https://doi.org/10.1016/j.jeem.2008.01.001>
- Hilborn R. Managing fisheries is Managing people: what has been learned? *Fish Fish* 2007;**8**:285–96. https://doi.org/10.1111/j.1467-2979.2007.00263_2.x
- Holland DS, Sanchirico J, Johnston R *et al*. *Economic Analysis for Ecosystem-Based Management*. Washington D.C.: RFF Press. 2010
- Holland DS, Sutinen JG. Location choice in New England trawl fisheries: old habits die hard. *Land Econ* 2000;**76**:133. <https://doi.org/10.2307/3147262>
- Huang L, Smith MD. The dynamic efficiency costs of common-pool resource exploitation. *Am Econ Rev* 2014;**104**:4071–103. <https://doi.org/10.1257/aer.104.12.4071>
- Hunt LM, Boots BN, Boxall PC. Predicting fishing participation and site choice while accounting for spatial substitution, trip timing, and trip context. *North American J Fisheries Manage* 2007;**27**:832–47. <https://doi.org/10.1577/M06-079.1>
- ICES. *Strategic Plan*. 2021 <https://doi.org/10.17895/ices.pub.7460>
- Klaiber HA, von Haefen RH. Do random coefficients and alternative specific constants improve policy analysis? An empirical investigation of model fit and prediction. *Environ Resource Econ* 2019;**73**:75–91. <https://doi.org/10.1007/s10640-018-0250-z>
- Klier T, McMillen DP. Clustering of auto supplier plants in the United States: generalized method of moments spatial logit for large samples. *J Bus Econom Statist* 2008;**26**:460–71. <https://doi.org/10.1198/073500107000000188>
- Kristensen K. RTMB: “R” Bindings for “TMB.” R Package Version 1.8. Released. 2025
- Krisztin T, Piribauer P, Wögerer M. A spatial multinomial logit model for analysing urban expansion. *Spatial Econ Anal* 2022;**17**:223–44. <https://doi.org/10.1080/17421772.2021.1933579>
- McFadden D. The measurement of urban travel demand. *J Public Econ* 1974;**3**:303–28. [https://doi.org/10.1016/0047-2727\(74\)90003-6](https://doi.org/10.1016/0047-2727(74)90003-6)
- Parsons GR, Kealy MJ. Randomly drawn opportunity sets in a random utility model of Lake Recreation. *Land Econ* 1992;**68**:93. <https://doi.org/10.2307/3146746>
- Peterson G. Political ecology and ecological resilience: an integration of human and ecological dynamics. *Ecol Econ* 2000;**35**:323–36. [https://doi.org/10.1016/S0921-8009\(00\)00217-2](https://doi.org/10.1016/S0921-8009(00)00217-2)
- Pfeiffer L, Gratz T. The effect of rights-based fisheries management on risk taking and fishing safety. *Proc Natl Acad Sci* 2016;**113**:2615–20. <https://doi.org/10.1073/pnas.1509456113>
- R Core Team (2024). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>.
- Reimer MN, Abbott JK, Haynie AC. Structural behavioral models for rights-based fisheries. *Resour Energy Econ* 2022;**68**:101294. <https://doi.org/10.1016/j.reseneeco.2022.101294>
- Reimer MN, Abbott JK, Wilen JE. Fisheries production: management institutions, spatial choice, and the quest for policy invariance. *Mar Resour Econ* 2017;**32**:143–51. <https://doi.org/10.1086/690678>
- Revelt D, Train K. Mixed logit with repeated choices: household's choices of appliance efficiency level. *Rev Econ Stat*. 1998;**80**:647–57. <https://doi.org/10.1162/003465398557735>
- Schmidheiny K, Brülhart M. On the equivalence of location choice models: conditional logit, nested logit and poisson. *J Urban Econ* 2011;**69**:214–22. <https://doi.org/10.1016/j.jue.2010.09.004>
- Smith MD, Wilen JE. Economic impacts of marine reserves: the importance of spatial behavior. *J Environ Econ Manag* 2003;**46**:183–206. [https://doi.org/10.1016/S0095-0696\(03\)00024-X](https://doi.org/10.1016/S0095-0696(03)00024-X)
- Smith MD. State dependence and heterogeneity in fishing location choice. *J Environ Econ Manag* 2005;**50**:319–40. <https://doi.org/10.1016/j.jeem.2005.04.001>
- Stafford TM. Accounting for outside options in discrete choice models: an application to commercial fishing effort. *v* 2018;**88**:159–79. <https://doi.org/10.1016/j.jeem.2017.10.006>
- Tardiff TJ. UC Irvine Working Paper Series: 78-11. *The Use of Alternative Specific Constants in Choice Modeling*. 1978.
- Tra CI. A discrete choice equilibrium approach to valuing large environmental changes. *J Public Econ* 2010;**94**:183–96. <https://doi.org/10.1016/j.jpubeco.2009.10.006>

- Train K. *Discrete Choice Methods with Simulation*. Cambridge, United Kingdom: Cambridge University Press. 2002.
- Valcic B. Spatial policy and the behavior of fishermen. *Mar Policy* 2009;**33**:215–22. <https://doi.org/10.1016/j.marpol.2008.06.001>
- Vasquez Caballero S, Sylvia G, Holland DS. Fishery participation and location choice model: the West Coast salmon troll commercial Fishery. *Can J Fish AquatSci* 2023;**80**:1770–84. <https://doi.org/10.1139/cjfas-2023-0001>
- Ward JM, Sutinen JG. Vessel entry-exit behavior in the Gulf of Mexico shrimp fishery. *Am J Agric Econ* 1994;**76**:916–23. <https://doi.org/10.2307/1243751>
- Werner S, DePiper G, Di J *et al*. Estimation of commercial fishing trip costs using sea sampling data. *Mar Resour Econ* 2020;**35**:379–410. <https://doi.org/10.1086/710668>
- Wilén JE, Smith MD, Lockwood D *et al*. Avoiding surprises: incorporating fisherman behavior into management models. *Bull Mar Sci* 2002;**70**:553–75.
- Wu H-H, Thorson JT, Hilborn R *et al*. Spatially varying coefficients improve discrete choice models for tuna purse seine fisheries in the western–Central Pacific. *ICES J Mar Sci* 2025;**82**:1–16. <https://doi.org/10.1093/icesjms/fsaf114>

Handling editor: Sean Pascoe